JECRC UNIVERSITY JAIPUR

Session: 2025–26

Subject: Mathematics-I (DMA024A)

Course: B. Tech - I Semester

Tutorial Sheet – 5

UNIT – 2 M.M: 64

CO2: To understand and apply concepts of **extreme values of functions** and **linear approximation**, which are useful in optimization and estimation problems in engineering applications.

PART – A (5 × 2 = 10 marks)

1. Define local maximum and local minimum of a function.
2. Write the necessary condition for extreme values of a differentiable function.
3. State the second derivative test for extreme values.
4. Write the formula of linear approximation for a function $f(x)$ near $x = a$.
5. Define error and approximate error in linear approximation.

PART – B (3 × 7 = 21 marks)

1. Find the maximum and minimum values of $f(x) = x^3 - 6x^2 + 9x + 1$.
2. Show that the function $f(x) = \cos x + \sin x$ attains its maximum and minimum values at suitable points.
3. Use linear approximation to estimate $\sqrt{10}$, taking $a = 9$.

PART – C (3 × 11 = 33 marks)

1. Find the maximum and minimum values of $f(x) = x^3 - 18x^2 + 96x$ and determine their nature.
2. A cube has edge length measured as 10 cm with an error of 0.02 cm. Use linear approximation to estimate the error in the volume of the cube.
3. Find the extreme values of $f(x) = x^4 - 8x^2 + 18$. Determine whether they are maxima or minima.

Answer Key:Part B

1. Local maximum at (1, 5) and local minimum at (3, 1).
2. Maximum value = $\sqrt{2}$ at $x = \pi/4$, minimum value = $-\sqrt{2}$ at $x = 5\pi/4$.
3. $\sqrt{10} \approx 3.1667$ (true value ≈ 3.1623).

Part C

1. Minimum at x=8 and maximum at x=4.
2. Estimated error in volume = 6 cm^3.
3. Extreme values at x=0 and x=±2. Local maximum $f(0) = 18$, local minima $f(\pm 2) = 2$.